

Data Types, Variables, Operators

Study Guide

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2.3.1 Type Conversion in Java

Type Conversion refers to changing the data type of a value or variable from one type to another. This is essential in Java when:

- Assigning a value of one data type to another
- Performing operations involving multiple data types

◆ Types of Type Conversion in Java

Java supports **two main types** of type conversion:

✓ A. Implicit Type Conversion (Widening Conversion)

Definition: Automatic type conversion performed by Java when the source data type can safely fit into the target data type (no data loss).

🔑 Key Points:

- Happens automatically.
- Safe and no risk of data loss.
- Done when converting from smaller to larger data types.

🔄 Common Conversion Path:

byte → short → int → long → float → double

char → int → long → float → double

✓ Example:

```
int num = 100;  
double result = num; // int to double  
System.out.println(result); // Output: 100.0
```

✓ B. Explicit Type Conversion (Type Casting / Narrowing Conversion)

Definition: Manual type conversion where a larger data type is converted to a smaller data type using a cast operator.

Key Points:

- Needs casting syntax.
- May lead to data loss or overflow.
- Used when narrowing the type (e.g., double to int).

Syntax:

```
dataType variableName = (targetType) value;
```

Example:

```
double pi = 3.14;
int approx = (int) pi; // double to int
System.out.println(approx); // Output: 3
```

2.3.2 Casting Between Primitive Types

Allowed Casts:

From Type → To Type	Casting Required?	Risk
int → long	✗ No	✗ No
long → float	✗ No	✗ No
double → int	✓ Yes	✓ Truncation
char → int	✗ No	✗ No
int → char	✓ Yes	✓ Overflow risk

2.3.3 Memory Behavior of All Data Types

Data Type	Memory Used	Implicit Cast Possible?
byte	1 byte	→ short, int, long, float, double
short	2 bytes	→ int, long, float, double
int	4 bytes	→ long, float, double
long	8 bytes	→ float, double
float	4 bytes (less precision)	→ double
double	8 bytes	✗ Can't cast automatically to smaller types
char	2 bytes	→ int, long, float, double

➤ Summary of Key Concepts:

Type Conversion changes a variable's data type.

- ◆ **Implicit Conversion (Widening)**
 - Done automatically by Java
 - Safe, no data loss
 - Example: int → double
- ◆ **Explicit Conversion (Narrowing / Casting)**
 - Done manually using (type)
 - May cause data loss or overflow
 - Example: double d = 9.8; int i = (int) d;
- ◆ **Key Rule:**
 - Small → Big: ✓ Automatic
 - Big → Small: ⚠ Needs casting, may lose data

➤ References:

☐ Official Java Documentation by Oracle

[Java Language and Virtual Machine Specifications – Oracle Docs](#)

- Provides specifications on type conversions and casting rules in Java.

☐ Oracle Java Tutorials

[The Java™ Tutorials - Data Types](#)

- Covers primitive data types, type promotion, and casting.

☐ GeeksforGeeks

Type Conversion in Java

- Beginner-friendly explanations and examples of both implicit and explicit conversions.

☐ W3Schools

Java Type Casting

- Easy-to-understand examples and concise descriptions for learners.

☐ TutorialsPoint

Java - Type Casting

- Good for quick revision and additional examples.

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